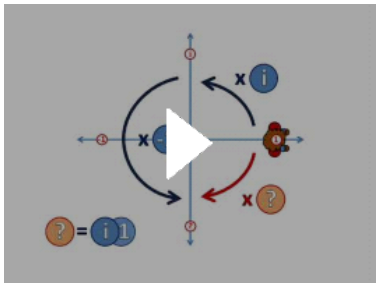
Day 28 HW

Monday, May 1, 2023 8:19 PM

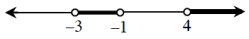
Polynomials	5 Problems

[An easy, alternative introduction to Imaginary Numbers]

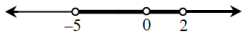


8-31.
You can use a number line to represent the x-values for which a polynomial graph is above or below the x-axis. The bold parts of each number line below show where the output values of a polynomial function are positive. That is, where the graph is above the x-axis. The open circles show locations of the x-intercepts or roots of the function. Where there is no shading, the value of the function is negative. Sketch a possible graph to fit each number line, and then write a possible equation. Each number line represents the x-axis for a different polynomial.

a.



b.



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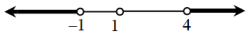
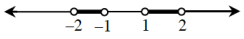


d.

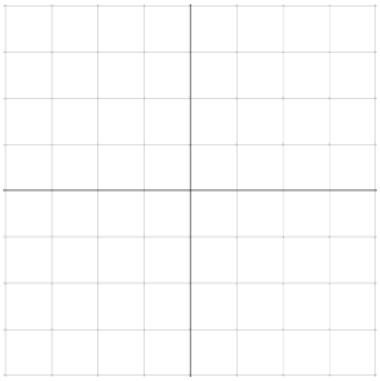


8-33.
Choose three of the polynomials you graphed in the Polynomial Functions Investigation (problem 8-2) and create number lines for their graphs similar to the ones in problem 8-31.

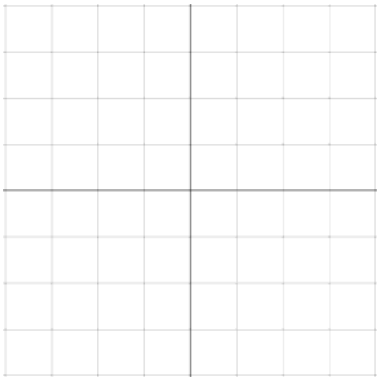
$P_1(x) = (x-2)(x+5)^2$	$P_2(x) = 2(x-2)(x+2)(x-3)$
$P_3(x) = x^4 - 21x^2 + 20x$	$P_4(x) = (x+3)^2(x+1)(x-1)(x-5)$
$P_5(x) = -0.1x(x+4)^3$	$P_6(x) = x^4 - 9x^2$
$P_7(x) = 0.2x(x+1)(x-3)(x+4)$	$P_8(x) = x^4 - 4x^3 - 3x^2 + 10x + 8$

8-35.
Without using a calculator, sketch rough graphs of the following functions.

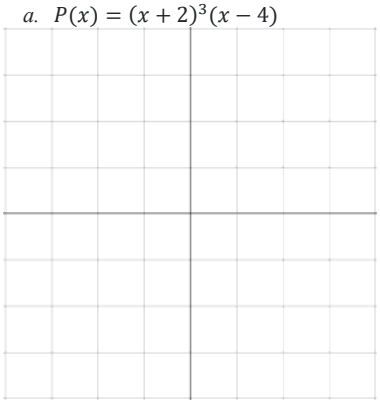
a. $P(x) = -x(x+1)(x-3)$



a. $P(x) = (x-1)^2(x+2)(x-4)$



a. $P(x) = (x+2)^3(x-4)$



8-36.
Where does the graph of $y = (x+3)^2 - 5$ cross the x-axis? Homework Help ✎

8-39.
What is the degree of each polynomial function below? Homework Help ✎

a. $P(x) = 0.08x^2 + 28x$

b. $y = 8x^2 - \frac{1}{7}x^5 + 9$

c. $f(x) = 5(x+3)(x-2)(x+7)$

d. $y = (x-3)^2(x+1)(x^3+1)$

Solutions: Remember these are just answers to check your work with. You must show your reasoning to receive full credit.

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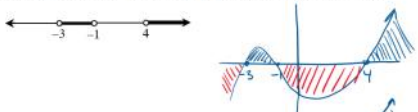
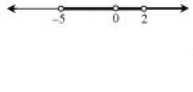
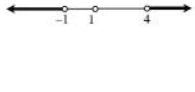
8-31.
You can use a number line to represent the x-values for which a polynomial graph is above or below the x-axis. The bold parts of each number line below show where the output values of a polynomial function are positive. That is, where the graph is above the x-axis. The open circles show locations of the x-intercepts or roots of the function. Where there is no shading, the value of the function is negative. Sketch a possible graph to fit each number line, and then write a possible equation. Each number line represents the x-axis for a different polynomial.

a. 

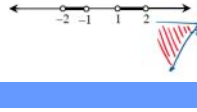
$y = (x+3)(x+1)(x-4)$

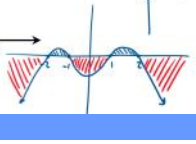
b. 

$y = x^2(x+5)(x-2)^2$

c. 

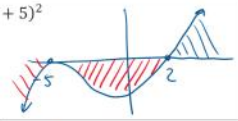
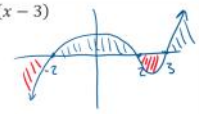
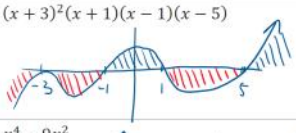
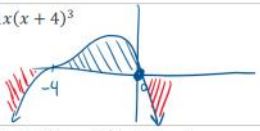
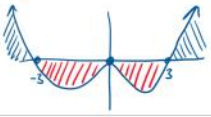
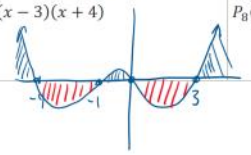
$y = (x+1)(x-1)^2(x-4)$

d. 



$y = -(x+2)(x-1)(x+1)(x-2)$

8-33.
Choose three of the polynomials you graphed in the Polynomial Functions Investigation (problem 8-2) and create number lines for their graphs similar to the ones in problem 8-31.

$P_1(x) = (x-2)(x+5)^2$ 	$P_2(x) = 2(x-2)(x+2)(x-3)$ 
$P_3(x) = x^4 - 21x^2 + 20x$ $x(x^3 - 21x + 20)$ TRICKY	$P_4(x) = (x+3)^2(x+1)(x-1)(x-5)$ 
$P_5(x) = -0.1x(x+4)^3$ 	$P_6(x) = x^4 - 9x^2$ $x^2(x^2-9)$ $x^2(x+3)(x-3)$ 
$P_7(x) = 0.2x(x+1)(x-3)(x+4)$ 	$P_8(x) = x^4 - 4x^3 - 3x^2 + 10x + 8$ $x^3(x-4) - (3x^2 - 10x - 8)$ $x^3(x-4) - (3x+2)(x-4)$ $(x-4)(x^3 - 3x + 2)$ TRICKY

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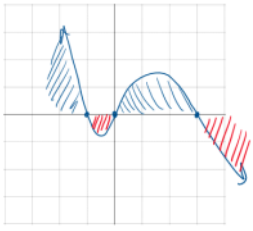
a. $P(x) = -x(x+1)(x-3)$



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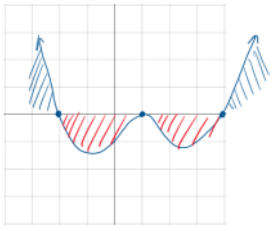

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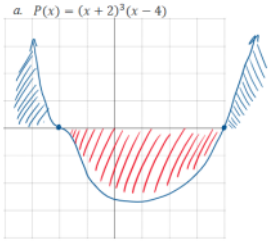

8-36.
Where does the graph of $y = (x+3)^2 - 5$ cross the x-axis? Homework Help

$y = (x+3)(x+3) - 5$
$y = x^2 + 6x + 9 - 5$
$y = x^2 + 6x + 4$
$0 = x^2 + 6x + 4$

$x = \frac{-6 \pm \sqrt{36 - 4(1)(4)}}{2}$

$x = \frac{-6 \pm \sqrt{20}}{2}$

$x = \frac{-6 \pm 2\sqrt{5}}{2} = -3 \pm \sqrt{5}$

$(-3+\sqrt{5}, 0)$ & $(-3-\sqrt{5}, 0)$

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What is the degree of each polynomial function below? Homework Help

a. $P(x) = 0.08x^2 + 28x$ Degree 2

b. $y = 8x^2 - \frac{1}{7}x^5 + 9$ Degree 5

c. $f(x) = 5(x+3)(x-2)(x+7)$ Degree 3

d. $y = (x-3)^2(x+1)(x^3+1)$ Degree 6